Experiment 6

Inference of Q Factor of Rectangular Cavity Resonator for TE101 Mode with Dimensions a, b and c

Test Case 1

A rectangular cavity resonator operating in the TE101 mode is made of copper ($\sigma = 5.8\times10^7$ S/m). Starting from a = 4 cm, b = 3 cm and varying the cavity length c from 8 cm to 12 cm, determine how the Q-factor changes and explain the relationship between the dimension c and the Q-factor.

Code -

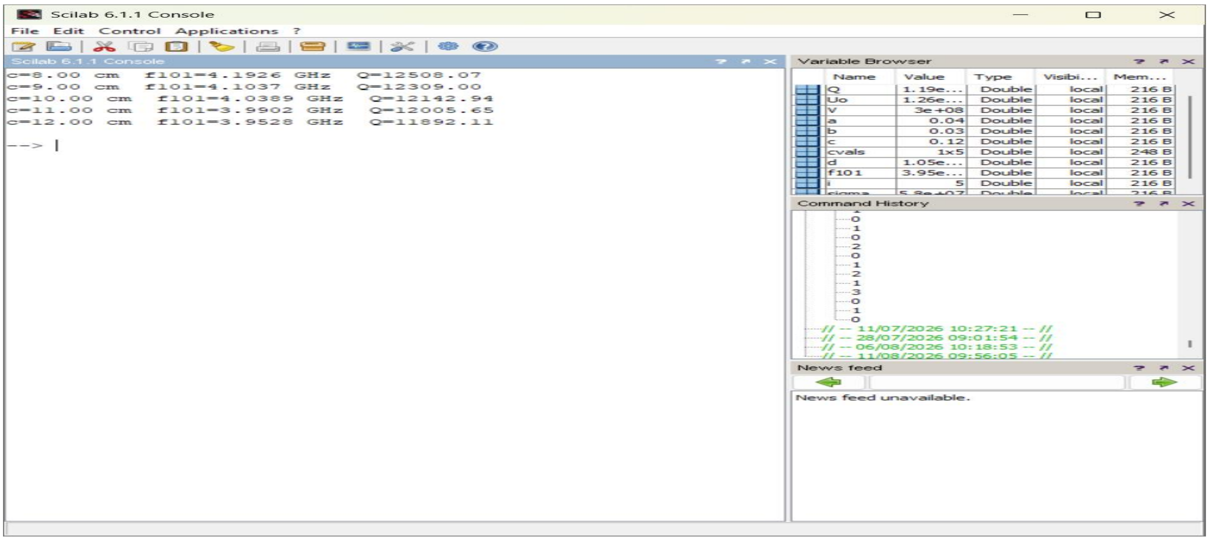
```
clear; clc;
sigma=5.8*10^7; Uo=4*%pi*10^-7; V=3*10^8;
a=0.04; b=0.03;
cvals=[0.08 0.09 0.10 0.11 0.12];
for i=1:5
  c=cvals(i);
  f101=(V/2)*sqrt((1/a)^2+(1/c)^2);
  d=sqrt(1/(%pi*f101*Uo*sigma));
  Q=(a^2+c^2)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));
  printf('c=%.2f cm  f101=%.4f GHz  Q=%.2f\n',c*100,f101*10^-9,Q);
end
```

Output:

Fixed: a = 4.0 cm, b = 3.0 cm (Copper, $\sigma = 5.8\times10^7$ S/m)

a (cm)	b (cm)	c (cm)	f101 (GHz)	Q Factor - Sim	Q Factor - Theory
4.0	3.0	8.0	4.1926	12508.07	12508.07

4.0	3.0	9.0	4.1037	12309.00	12309.00
4.0	3.0	10.0	4.0389	12142.94	12142.94
4.0	3.0	11.0	3.9902	12005.65	12005.65
4.0	3.0	12.0	3.9528	11892.11	11892.11



Result:

As the cavity length c is increased from 8 cm to 12 cm (with a = 4 cm, b = 3 cm held fixed), the resonant frequency f101 decreases and the Q-factor steadily decreases from 12508 to 11892. This is because increasing c lowers the surface-to-volume ratio efficiency of the TE101 mode and reduces the resonant frequency, both of which reduce the stored-to-dissipated energy ratio represented by Q.

Test Case 2

Using the dimension sets from Test Case 1, determine which combination of a, b and c yields the maximum Q-factor and explain why.

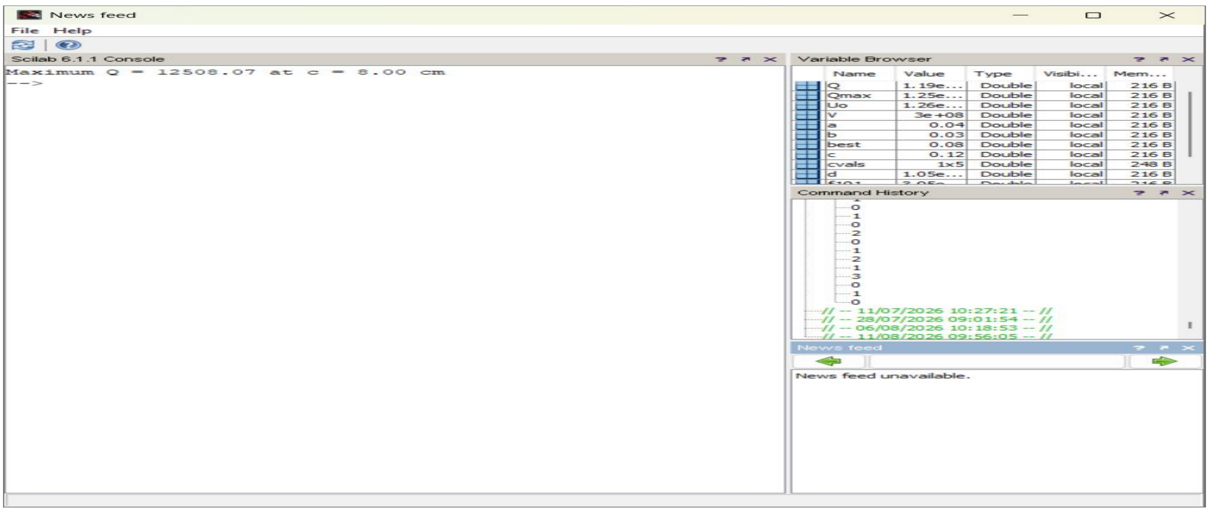
Code -

```
clear; clc;
sigma=5.8*10^7; Uo=4*%pi*10^-7; V=3*10^8;
a=0.04; b=0.03;
```

```
cvals=[0.08 0.09 0.10 0.11 0.12];
Qmax=0; best=0;
for i=1:5
  c=cvals(i);
  f101=(V/2)*sqrt((1/a)^2+(1/c)^2);
  d=sqrt(1/(%pi*f101*Uo*sigma));
  Q=(a^2+c^2)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));
  if Q>Qmax then Qmax=Q; best=c; end
end
printf('Maximum Q = %.2f at c = %.2f cm',Qmax,best*100);
```

Output:

a (cm)	b (cm)	c (cm)	Q Factor - Simulation	Q Factor - Theoretical
4.0	3.0	8.0	12508.07	12508.07
4.0	3.0	9.0	12309.00	12309.00
4.0	3.0	10.0	12142.94	12142.94
4.0	3.0	11.0	12005.65	12005.65
4.0	3.0	12.0	11892.11	11892.11



Result:

Among the tested dimension sets, the combination a = 4 cm, b = 3 cm, c = 8 cm gives the highest Q-factor (12508.07). The smallest cavity length c gives the highest resonant frequency and the most favourable ratio of stored volume to dissipating wall surface area for the TE101 mode, resulting in the maximum Q-factor.

Test Case 3

Fix b = 4 cm and c = 10 cm, and vary a between 4 cm and 6 cm. Determine how the Q-factor changes and comment on the sensitivity of Q to the dimension a.

Code -

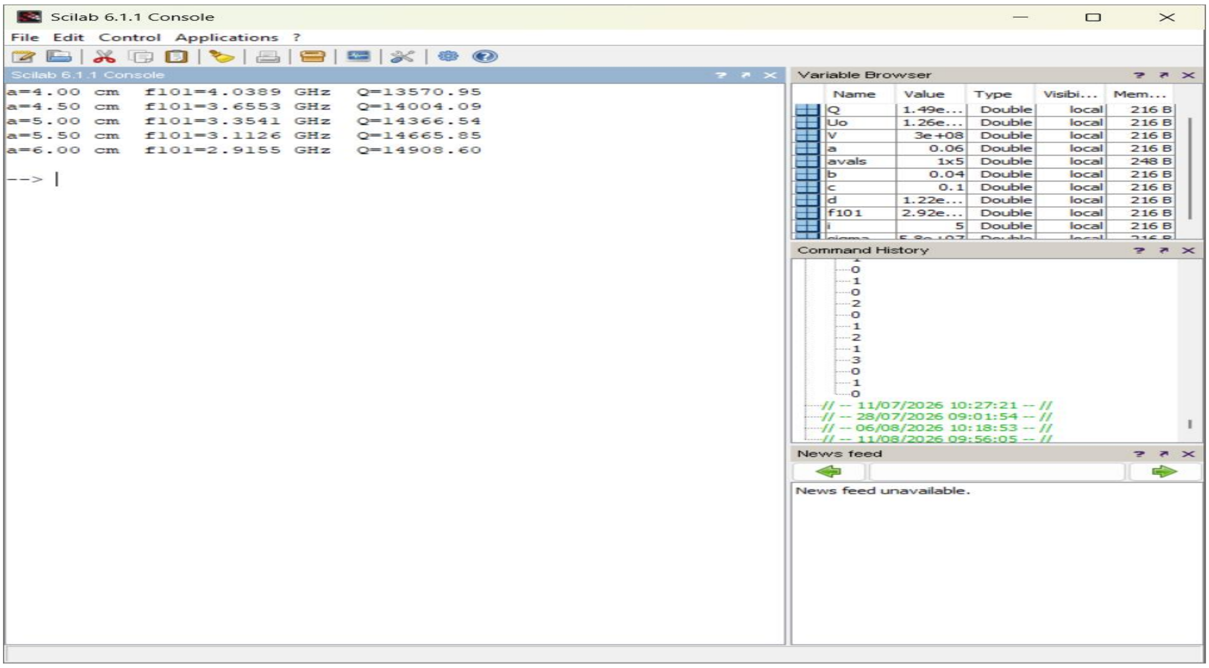
```
clear; clc;
sigma=5.8*10^7; Uo=4*%pi*10^-7; V=3*10^8;
b=0.04; c=0.10;
avals=[0.040 0.045 0.050 0.055 0.060];
for i=1:5
  a=avals(i);
  f101=(V/2)*sqrt((1/a)^2+(1/c)^2);
  d=sqrt(1/(%pi*f101*Uo*sigma));
  Q=(a^2+c^2)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));
  printf('a=%.2f cm  f101=%.4f GHz  Q=%.2f\n',a*100,f101*10^-9,Q);
end
```

Output:

Fixed: b = 4.0 cm, c = 10.0 cm

a (cm)	b (cm)	c (cm)	f101 (GHz)	Q Factor - Sim	Q Factor - Theory
4.0	4.0	10.0	4.0389	13570.95	13570.95
4.5	4.0	10.0	3.6553	14004.09	14004.09

5.0	4.0	10.0	3.3541	14366.54	14366.54
5.5	4.0	10.0	3.1126	14665.85	14665.85
6.0	4.0	10.0	2.9155	14908.60	14908.60



Result:

As a increases from 4 cm to 6 cm (b = 4 cm, c = 10 cm fixed), the Q-factor rises steadily from 13571 to 14909, although the resonant frequency drops. The rate of increase in Q flattens slightly at larger a, showing that Q is moderately sensitive to a but with diminishing returns. In practical cavity design, the dimension along which the dominant field variation occurs (here a, in the x-direction for TE101) is usually the most critical one to optimise for achieving a high Q-factor, since it directly sets the mode's resonant condition and its associated wall-loss surface area.

Test Case 4

Fix a = 5 cm and c = 10 cm, and vary b (the y-axis dimension) between 3 cm and 5 cm. Analyze how the Q-factor changes with b.

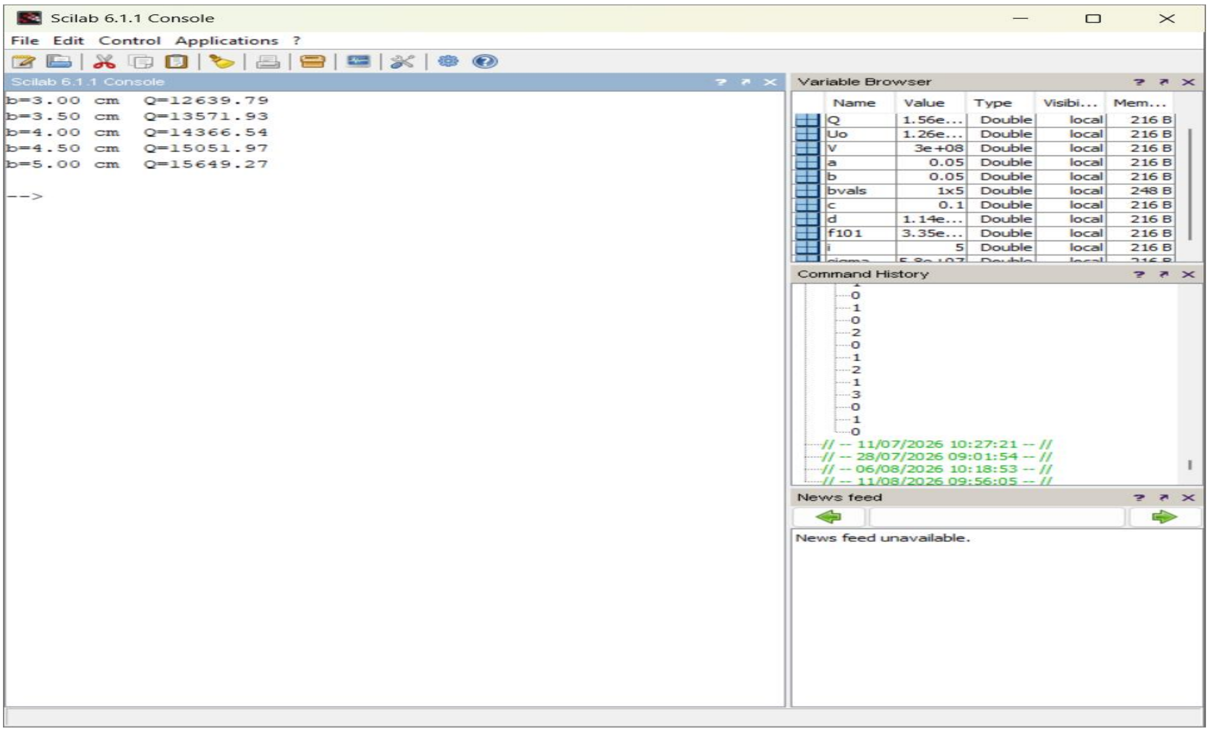
Code -

```
clear; clc;
sigma=5.8*10^7; Uo=4*%pi*10^-7; V=3*10^8;
a=0.05; c=0.10;
bvals=[0.030 0.035 0.040 0.045 0.050];
for i=1:5
  b=bvals(i);
  f101=(V/2)*sqrt((1/a)^2+(1/c)^2);
  d=sqrt(1/(%pi*f101*Uo*sigma));
  Q=(a^2+c^2)*a*b*c/(d*(2*b*(a^3+c^3)+a*c*(a^2+c^2)));
  printf('b=%.2f cm  Q=%.2f\n',b*100,Q);
end
```

Output:

Fixed: a = 5.0 cm, c = 10.0 cm (f101 = 3.3541 GHz, independent of b for TE101)

a (cm)	b (cm)	c (cm)	Q Factor - Simulation	Q Factor - Theoretical
5.0	3.0	10.0	12639.79	12639.79
5.0	3.5	10.0	13571.93	13571.93
5.0	4.0	10.0	14366.54	14366.54
5.0	4.5	10.0	15051.97	15051.97
5.0	5.0	10.0	15649.27	15649.27



Result:

Increasing b from 3 cm to 5 cm (with a = 5 cm, c = 10 cm fixed) raises the Q-factor from 12640 to 15649, since the TE101 resonant frequency does not depend on b, so increasing b only adds cavity volume without adding additional conducting wall-loss area in the field-varying directions, improving the volume-to-loss ratio and hence the Q-factor.